AI IN RESEARCH

Lesson Plan

Audience: Upper-division political science research class
Duration: 90 minutes
Format: Hands-on workshop; bring laptops
Demo topic used throughout: Autonomous Weapons Systems (AWS)

Session goals

By the end of the session, students will be able to:

1. Use Gemini to **refine a broad research interest** into a focused, researchable question
2. Use NotebookLM to **build a small, source-grounded knowledge base** and generate an outline they can trust
3. Translate a refined question into **Boolean search strings** and test them in EBSCO Discovery Service (EDS)
4. Recognize three core risks of AI-assisted research: **unverified citations, hallucinations, and privacy**

Materials & prep

- Projector + instructor laptop signed in to: Google account (Gemini + NotebookLM), library proxy (EDS)
- **3–5 seed sources for the AWS demo topic** saved as PDFs / links on the instructor laptop (a law review article, an ICRC report, a policy brief, one academic book chapter) — loaded *during* Phase 2 so students see the full build process
- Slide deck (small — ~9 slides)
- Survey of student research interests (already collected — instructor will reference)
- One-page handout (optional): the workflow map + 3 critical-use cautions
- Backup: a screenshot of a "fabricated citation" failure example in case live demo doesn't produce one

Session at a glance

Time	Block	What happens
0:00–0:10	**Opening + framing**	Goals, the research workflow, 3 cautions
0:10–0:35	**Phase 1 — Refine the question (Gemini)**	Demo (12 min) → student hands-on (10 min) → quick share (3 min)
0:35–1:00	**Phase 2 — Outline + seed sources (NotebookLM)**	Guided live build — instructor and students build in parallel, step-for-step
1:00–1:20	**Phase 3 — Search strings (Gemini → EDS)**	Demo (10 min) → student hands-on (10 min)

Time	Block	What happens
1:20–1:30	**Wrap**	Workflow recap, takeaways, where to get help

0:00–0:10 — Opening + framing (10 min)

Talking points

- **Today is hands-on.** You will leave with: a refined research question, an outline grounded in real sources, and a working search string for EDS.
- **AI does not replace the research process. It accelerates it.** The process is the same one you've always used:
  - Question → Outline → Keywords → Search → Evaluate → Iterate
- **Where AI helps:** brainstorming, scoping, sub-question generation, vocabulary discovery, draft outlines, query construction.
- **Still predicting, now working in a loop.** These tools are prediction engines underneath. They write what usually comes next, not what they have checked. Newer ones like Gemini Deep Research add a loop. They plan, search, read what they find, back up from dead ends, and try again. That makes them more persistent, not more truthful, which is why the cautions below still apply.
- **Where AI hurts (the 3 cautions):**
  - **Unverified citations.** A general LLM will invent citations that look real. Always click through.
  - **Hallucination of facts.** Sounds confident, can be wrong. Cross-check with sources.
  - **Privacy / what to upload.** Don't upload copyrighted PDFs you don't own, unpublished work, personal data, or anything sensitive — even into "private" notebooks.
- **Today's pattern:** I'll work through each step on one shared topic (Autonomous Weapons Systems); you apply the same step to *your* topic from the survey.

Instructor note

Keep this block tight. The point is to set up the workflow; the depth comes in the hands-on phases.

0:10–0:35 — Phase 1: Refine the question with Gemini (25 min)

Phase 1 goal

Move from a broad interest to a researchable, arguable question with named sub-questions and key vocabulary.

Phase 1 demo (12 min) — Autonomous Weapons Systems

Starting prompt (broad student-style question):

> *"I want to write about whether autonomous weapons systems are too dangerous and whether their effectiveness in wars justifies their existence."*

Demo these prompting moves in order:

5. **Clarify scope.**

> *"Help me narrow this. What are 4–5 different angles a political science paper could take on autonomous weapons systems? For each, give a one-sentence thesis and what discipline (IR, ethics, law, policy) it would draw on."*

1. **Pick an angle and stress-test it.**

> *"I want to focus on the international humanitarian law angle. What are the strongest counter-arguments to the claim that lethal autonomous weapons should be banned under IHL?"*

2. **Generate sub-questions.**

> *"Give me 5 sub-questions that, if answered, would let me argue this thesis well."*

3. **Extract vocabulary** (this feeds Phase 3).

> *"List the key terms, synonyms, and disciplinary jargon a librarian would use to search this topic in a political science database. Group them by concept."*

Phase 1 teaching points

- Notice Gemini suggested several angles — **you choose**, not Gemini
- The vocabulary list is doing real librarian work — synonyms are how you broaden a search
- **Caution moment:** if Gemini mentions specific scholars or cases, *do not* trust those yet — that's a Phase 2 (NotebookLM) job

Phase 1 student hands-on (10 min)

Each student takes their own topic from the survey and walks through the same 4 prompts in Gemini. End the block with:

- A refined research question (written down)
- A list of 5 sub-questions
- A vocabulary list grouped by concept

Phase 1 share-out (3 min)

2–3 students name their refined question and one surprising sub-question Gemini surfaced.

0:35–1:00 — Phase 2: Outline with NotebookLM (25 min)

Phase 2 goal

Convert a refined question + a few real sources into a **grounded, citation-backed outline** — and feel the difference between a general LLM and a sourced LLM.

Format: guided live build (everyone builds in parallel)

This phase is **not** demo-then-practice. The instructor builds the AWS notebook live; students build their own notebook on their topic in parallel, step-for-step. Pace is set by the slowest reasonable student; pause at each step to check the room.

Step-by-step (≈22 min, with the last few min for the contrast moment)

Step 1 — Create the notebook (≈2 min)

- Instructor: opens NotebookLM, creates a new notebook titled "Autonomous Weapons Systems."
- Students: create their own notebook titled with their refined Phase 1 question.

Step 2 — Talk about seed sources before adding any (≈3 min)

- What makes a good seed source: peer-reviewed article, policy/legal report, book chapter, established think-tank brief. Avoid blogs and AI-generated summaries.
- **Privacy beat:** only add what you have a right to use; don't upload paywalled PDFs you don't own.

Step 3 — Add 3–5 sources, narrating choices (≈6 min)

- Instructor: adds the AWS sources one at a time, narrating *why* each one is a good seed (a law review article for the legal frame, an ICRC report for the humanitarian frame, a policy brief for current debate, etc.).
- Students: add 2–3 sources for their own topic. If they don't have any yet, they can do a quick EDS search now (this previews Phase 3) or use a source they already had from class.
- Show the source types NotebookLM accepts: PDFs, links, Google Docs, pasted text, YouTube.

Step 4 — Generate a Briefing Doc / synthesis (≈2 min)

- Instructor runs it on the AWS sources; students run it on theirs.
- Point out: this is synthesis grounded in *your* sources, not general training data.

Step 5 — Ask an analytical question with citations (≈3 min)

- Instructor prompts NotebookLM: *"What are the main IHL objections to lethal autonomous weapons, according to my sources?"*
- Students ask one analytical question of their own notebook.
- Everyone clicks an inline citation and inspects the actual source passage. **This is the safety property:** NotebookLM cannot cite what isn't in the notebook.

Step 6 — Generate an outline (≈3 min)

- Instructor prompts: *"Draft a 5-section paper outline arguing that lethal autonomous weapons should be restricted under IHL. Ground each section in my sources."*
- Students generate their own outline.
- Quick discussion: where does the outline match your mental model? Where does it surface something you hadn't considered?

Step 7 — Bonus features, briefly (≈2 min, optional)

- Study guide / Q&A — useful for understanding the sources before writing.
- Audio overview — quick demo so they know it exists.

Phase 2 contrast moment — the critical-use beat (≈3 min)

- Side-by-side: ask Gemini the *same* analytical question and request citations.
- Show how Gemini's citations may be invented or vague ("a 2021 Stanford paper" with no link).
- Headline: **NotebookLM cannot cite what isn't in the notebook** — that's why it's the synthesis tool, while Gemini is the ideation tool.

Phase 2 wrap

One quick prompt to the room: *"Is your NotebookLM outline closer to a real paper outline than what you started with?"* — show of hands, move on.

1:00–1:20 — Phase 3: Search strings + EBSCO Discovery Service (20 min)

Phase 3 goal

Translate the Phase 1 vocabulary list into a working Boolean search string and iterate it against EDS.

Phase 3 demo (10 min) — Autonomous Weapons Systems

4. **Ask Gemini to draft Boolean variants.**

> *"Using these concept groups [paste vocabulary from Phase 1], draft 3 Boolean search strings for a political science database. Use AND between concepts, OR within a concept, parentheses to group, and truncation (*) where appropriate."*

5. **Pick one, paste into EDS.** Show the results count.
6. **Iterate live.**
   - Too many results → add another concept with AND, or narrow with a subject heading
   - Too few → broaden a synonym group, drop a concept, use truncation
   - Show **subject term suggestions** EDS surfaces and **field tags** (SU, TI, AB)
7. **Show source evaluation.** Open one result and walk through: peer-reviewed? Date? Discipline? Cited by? — fast triage.

Phase 3 teaching points

- **AI drafts the string; you refine it.** Gemini doesn't know which subject headings EDS actually uses. Real iteration happens in the database.
- **Optional aside:** for deeper poli-sci searching, also try native databases — *Worldwide Political Science Abstracts*, *PAIS Index*, *Academic Search Complete*. EDS is a great starting point, not the only one.

Phase 3 student hands-on (10 min)

Each student:

8. Asks Gemini to draft 2–3 Boolean strings from their Phase 1 vocabulary
9. Tests in EDS and iterates at least twice
10. Saves 2–3 promising results to add to their NotebookLM

1:20–1:30 — Wrap (10 min)

Workflow recap (the headline)

> *Gemini to refine → NotebookLM to ground → EDS to find → loop back.*

Each tool plays to its strength. The workflow is iterative — when Phase 3 surfaces a new concept, return to Phase 1 to add vocabulary, etc.

Three takeaways

11. **Verify every citation.** AI suggests; you verify.
12. **Use grounded tools (NotebookLM) for synthesis, general tools (Gemini) for ideation.**
13. **Be deliberate about what you upload.**

Where to get help

- Your librarian (the instructor) — office hours / appointment link
- Research guides for political science

- Library AI/research-tools guide (if one exists)

Optional exit ticket

14. What's your refined research question now?
15. What's one search string that worked?
16. What's one thing you'd still like help with?

Instructor checklist (day-of)

☐ AWS seed sources (3–5 PDFs/links) saved on the instructor laptop, ready to upload during Phase 2

☐ Signed into Gemini and NotebookLM on the projection laptop

☐ EDS open and authenticated (test from off-campus if relevant)

☐ Slide deck loaded

☐ Have a "fabricated citation" screenshot ready as a backup

☐ Have the student survey responses open for quick examples

☐ Optional handout printed

Stretch / fallback options

- **If running long:** skip Phase 2 Step 7 (bonus features); cut Phase 3 source-evaluation walk-through
- **If running short:** spend extra time in Phase 2 Step 7 with audio overview, or run a second EDS iteration in Phase 3
- **If the room's wifi struggles:** have screenshots of each step as a backup deck